

Katherine M. Yut

Geospatial Data Engineer

Expert in geospatial analysis, statistics, and programming using R and Python. Passionate about data visualization. Seeking to solve complex problems to make systems more efficient. Meticulous, intrinsically motivated, and a quick learner.

Work History

Grassroots Carbon, San Antonio, TX (hybrid)

Data Engineer 2025-08 – current

Building ETL/ELT data pipelines using Python, GitHub, and Docker containerization at a company passionate about doing right for the land, the rancher, and the world.

Skills: pydantic, Streamlit, cron, GitHub Actions, CI/CD, APIs, code review.

Grassroots Carbon, San Antonio, TX (hybrid)

Carbon Technical Manager 2024-08 – 2025-08

Automated rancher eligibility screening processes using R shiny and Python through the Salesforce API, provided geospatial data analysis using QGIS, and supported the navigation of carbon standards.

University of Texas, Bureau of Economic Geology, Austin, TX

GIS & Energy Data Analyst 2022-05 – 2024-07

Leveraged R to clean and manipulate data tables, solve problems, and automate processes; utilized ArcMap to aid in oil & gas research publications; and published research findings as papers, posters, and presentations.

Education

Master of Arts: Economics and Big Data

Bachelor of Arts: Economics and Big Data

Bachelor of Science: Geographic Information Science

University of Oklahoma (Norman, OK)

2016-08 – 2021-05, *Summa cum laude*

Research

KM Smye, **K Yut**, RC Reedy, BR Scanlon, JP Nicot, P Hennings (2024). Challenges with managing unconventional water production and disposal in the Permian Basin. AAPG Bulletin 108 (12), 2215-2240.

Peng, S., Maraggi, LMR., Bhattacharya, S., **Yut, K.**, McMahon, T., Haddad, M. (2023). Feasibility of CO2 storage in depleted unconventional oil and gas reservoirs: Capacity, microscale mechanism, injectivity, fault stability, and monitoring. URTEC 13–15 June 2023.

M Bull, T Francis, **K Yut**, K Weber, J Spruce, C Jarnevich. West Virginia Ecological Forecasting. NASA DEVELOP National Program, 2019.

Kelley, S., **Yut, K.**, Kulkarni, R., & Gaffin, D. (2019). Avoidance of rosemary oil by scorpions. Journal of Arachnology.

Contact

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Skills

Expert

RStudio, R, R shiny:

Writing functions, making packages, documenting code, creating interactive html servers.

ESRI ArcGIS Pro, QGIS:

Geospatial analyses, coordinate system management, map-making.

Creating timely deliverables and communicating effectively

Curiosity, organization, multitasking, and focus

Advanced

Python

GitHub

Salesforce backend

Big Data statistical processing, analysis, and modeling

Intermediate

SQL

Docker containerization

Google Earth Engine

ENVI